Internet Appendix to "The Pre-FOMC Announcement Drift"

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This appendix provides supplementary results. Section A details the bootstrap analyses discussed in Section III.D. of the paper. Section B provides additional Figures and Tables.

A. Bootstrap Analysis

A.1. Bootstrapped Standard Errors for Dummy Variable Regression

We assess the small-sample standard errors of the estimated constant and dummy variable coefficient in regression (1) with a simple bootstrap approach. More precisely, we draw with replacement a return series of length 131 from the observed distribution of the pre-FOMC returns in our 1994 to 2011 sample, and a series of length 4,010 from the observed distribution of non-FOMC returns. With the two artificial series at hand, we reestimate the dummy variable regression (1) and record the estimated coefficients. The empirical distribution of the estimated FOMC dummy coefficient across bootstrap replications is centered around a mean of 48.5 basis points with a standard deviation of 10.8 basis points (upper-left panel of Figure IA.1). These numbers are very close to the regression results in Table II and show that the asymptotic standard errors do not understate the sampling variability of the estimated means. For the 1980 to 2011 sample (upper-left panel of Figure IA.2) we also find bootstrapped coefficients that are very similar to our regression results in Table IV.

A.2. Random Sampling from Non-FOMC Return Distribution

One might ask how likely it is to observe an average of 49 basis points in a short random sample of non-FOMC day returns. We address this question in another bootstrap exercise by drawing with replacement from the distribution of non-FOMC returns a time series of length equal to the number of FOMC announcement days. The bootstrap distribution of sample averages for the 1994 to 2011 sample (upper-right panel of Figure IA.1) is centered around a mean of 0.7 basis points with a standard deviation of 10.5 basis points. More importantly, the probability of observing a mean greater than the 48.8 basis point sample average of pre-FOMC returns is zero. Repeating this exercise for the 1980-2011 period (upper-right panel of Figure IA.2), we find the same result. In summary, for both samples, it is essentially impossible that one could have observed such a large mean had one drawn from the distribution of returns outside the pre-FOMC window.

*Citation format: Lucca, David O. and Emanuel Moench, Internet Appendix to "The Pre-FOMC Announcement Drift," *Journal of Finance* [vol #], [pages], http://www.afajof.org/supplements.asp. Please note: Wiley-Blackwell is not responsible for the content or functionality of any supporting information supplied by the authors. Any queries (other than missing material) should be directed to the authors of the article.

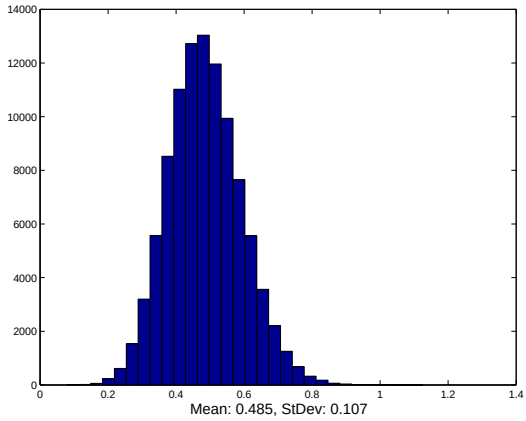
A.3. Random Sampling from Pseudo Pre-FOMC Return Distribution with Zero Mean

In light of the relatively small number of FOMC meetings and the high kurtosis of pre-FOMC day returns, one might be concerned that it is not unlikely to observe a sample mean as large as the documented pre-FOMC return in a short sample that is drawn from a distribution that has a population mean of zero. We assess this possibility in the following exercise. We construct a pseudo zero-mean pre-FOMC return distribution by subtracting the sample mean of 48.8 basis points from all 131 observations in the 1994 to 2011 sample. Thus, by construction, the resulting distribution has a sample mean of zero but identical higher moments as the observed distribution of pre-FOMC returns. We then randomly draw with replacement from the pseudo pre-FOMC distribution series of length 131. The histogram of sample means is shown in the lower-left panel of Figure IA.1. As expected, the distribution is tightly centered around zero. More importantly, we reject with a probability of 99.99% the possibility that we could have observed pre-FOMC returns with an average of 48.8 basis points if the true distribution had a population mean of zero. Repeating this exercise for the 1980 to 2011 sample (lower-left panel of Figure IA.2), we find that not a single series of draws has a mean greater than the 36.6 basis points that we observe for the pre-FOMC return over this period. Hence, we can safely rule out that the distribution of pre-FOMC returns has a population mean of zero.

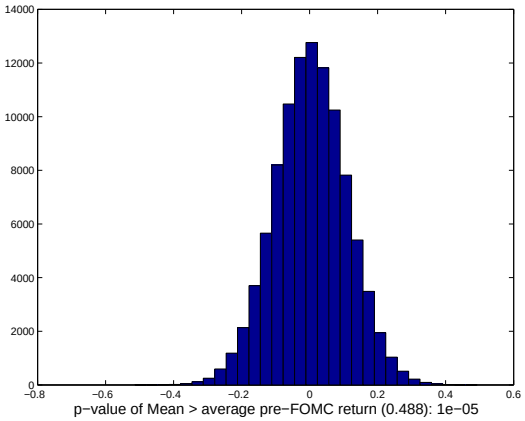
A.4. Data-Snooping

As previously discussed, we find the pre-FOMC returns to be highly statistically significant while the SPX does not display excess returns ahead of other major U.S. macroeconomic announcements. A skeptical reader may worry that the significance of our finding (and thus the Sharpe ratios) could be the artificial outcome of an extensive search across the universe of economic news announcements for the highest t-statistic. Of course, such a search would not bias the size of the return. We address this concern by carrying out a reality check in the spirit of White (2000). In particular, we simulate the snooping bias by resampling the 2pm-to-2pm excess return on the SPX and collecting the largest absolute t-statistic among the economic announcements considered in Table VII as well as FOMC announcements. The bootstrap distribution for the 1994 to 2011 sample (lower-right panel of Figure IA.1) shows that 99.98% of maximum absolute t-statistics are smaller than the value of 4.51 that we find for the pre-FOMC announcement return in the SPX for that sample (Table II, column (1)). Repeating this exercise for the 1980 to 2011 period (lower-right panel of Figure IA.2), we find an even smaller probability of having snooped a t-statistic larger than 4.86, which is the estimated asymptotic t-statistic of the pre-FOMC dummy for that sample (Table IV, last column). In other words, the statistical significance of our finding is extremely unlikely to be the result of a data-snooping exercise.

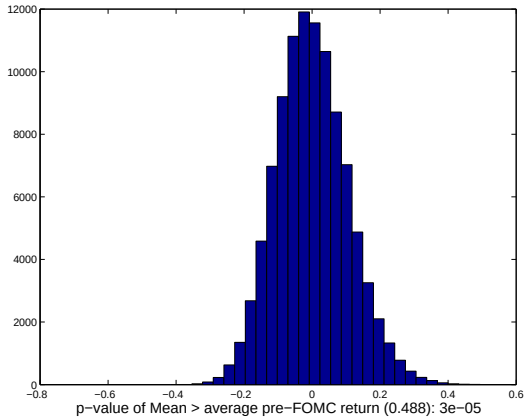
Bootstrapped Standard Errors for FOMC Dummy



Sampling from Non-FOMC Returns



Zero-Mean Pseudo Pre-FOMC Return Distribution



Data-Snooping

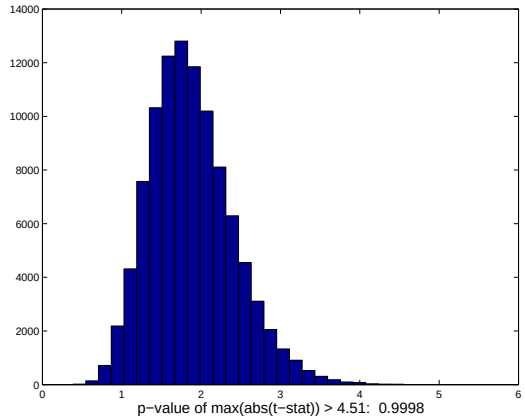
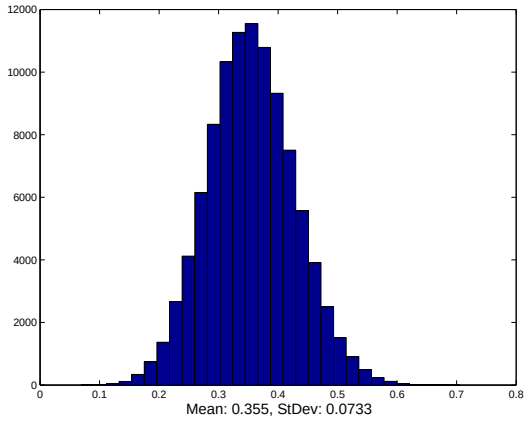
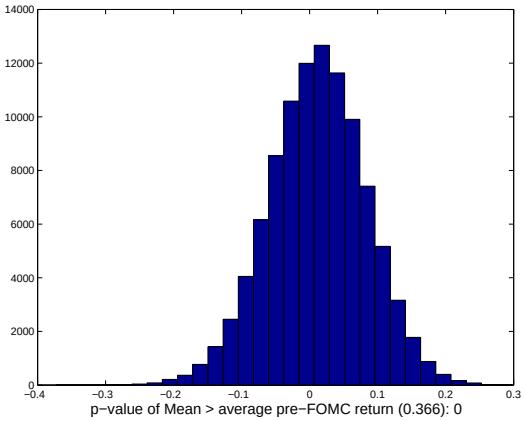


Figure IA.1. Bootstrap analysis: 1994-2011. This figure provides histograms of the bootstrap distributions discussed in Section III.D for the sample period September 1994 to March 2011.

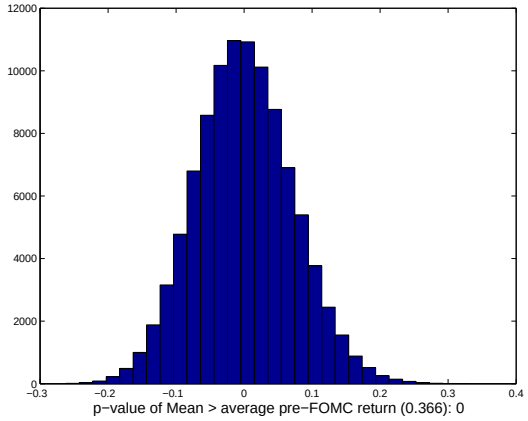
Bootstrapped Standard Errors for FOMC Dummy



Sampling from Non-FOMC Returns



Zero-Mean Pseudo Pre-FOMC Return Distribution



Data-Snooping

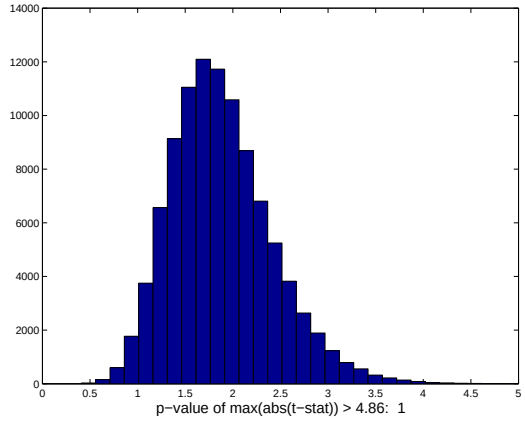


Figure IA.2. Bootstrap analysis: 1980-2011. This figure provides histograms of the bootstrap distributions discussed in Section III.D for the sample period January 1980 to March 2011.

B. Additional Figures and Tables

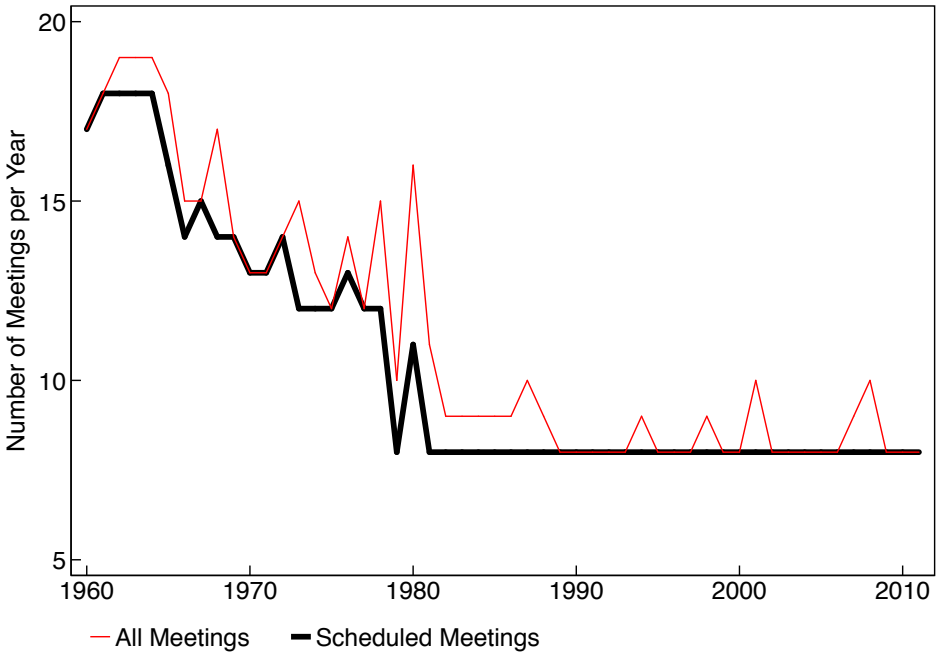


Figure IA.3. Number of FOMC meetings per year. This figure plots time series of the number of scheduled and the number of scheduled plus unscheduled FOMC meetings per year from 1960 to 2011.

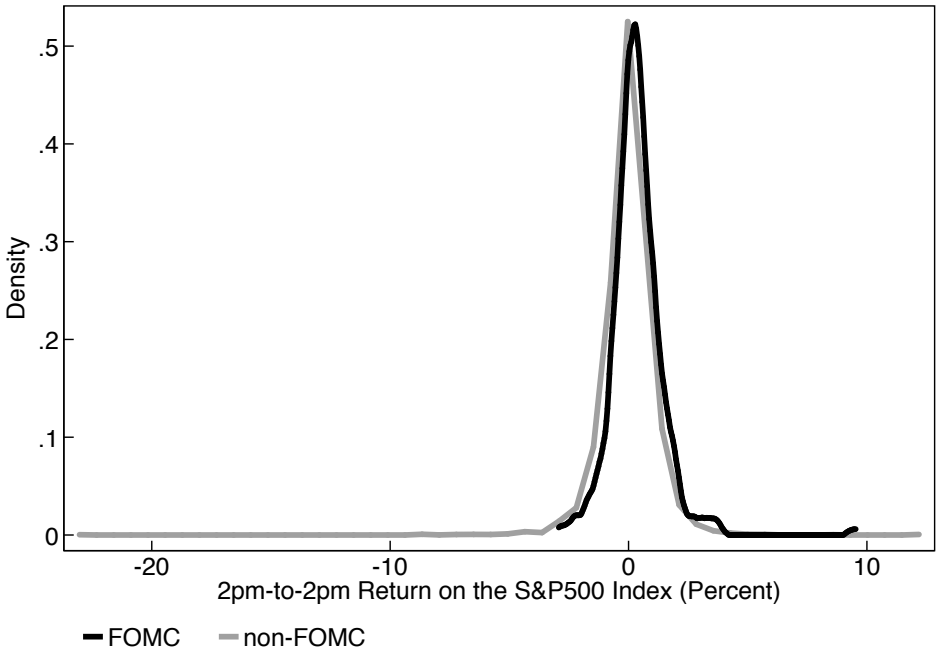


Figure IA.4. Empirical densities of pre-FOMC returns: 1980-2011. This figure plots empirical densities of the 2pm-to-2pm return on the SPX. The solid black line shows the return on days ahead of scheduled FOMC announcements and the gray line shows the return on all other days. The sample period is from January 1980 to March 2011.

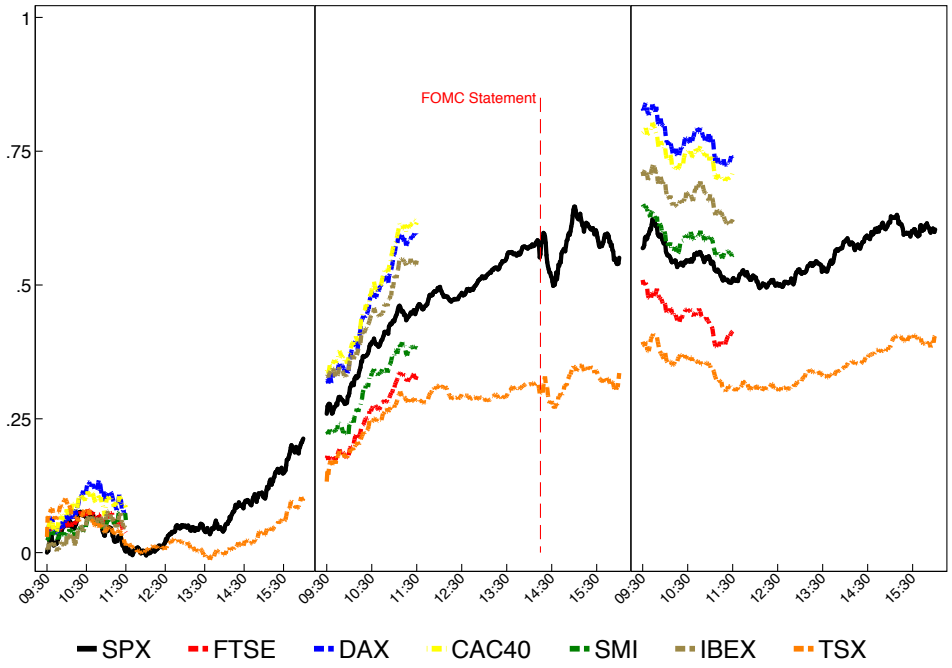


Figure IA.5. Cumulative returns on international stock market indices around FOMC announcements. This figure plots the average cumulative one-minute return on the SPX and other major international equity market indices over the three-day window around scheduled FOMC announcements. The solid black line shows the average cumulative return on the SPX from 9:30 am ET on the days before scheduled FOMC announcements until 4:00p.m. on days after scheduled FOMC announcements. The colored dashed lines show the cumulative returns on the German DAX, the British FTSE100, the French CAC40, the Spanish IBEX, the Swiss SMI, and the Canadian TSX over the same three-day window. All stock indices are shown only during hours of trading on the respective exchanges. The sample period is from January 1996 to March 2011. The dashed vertical line is set at 2:15pm ET, when FOMC announcements were typically released during that period.

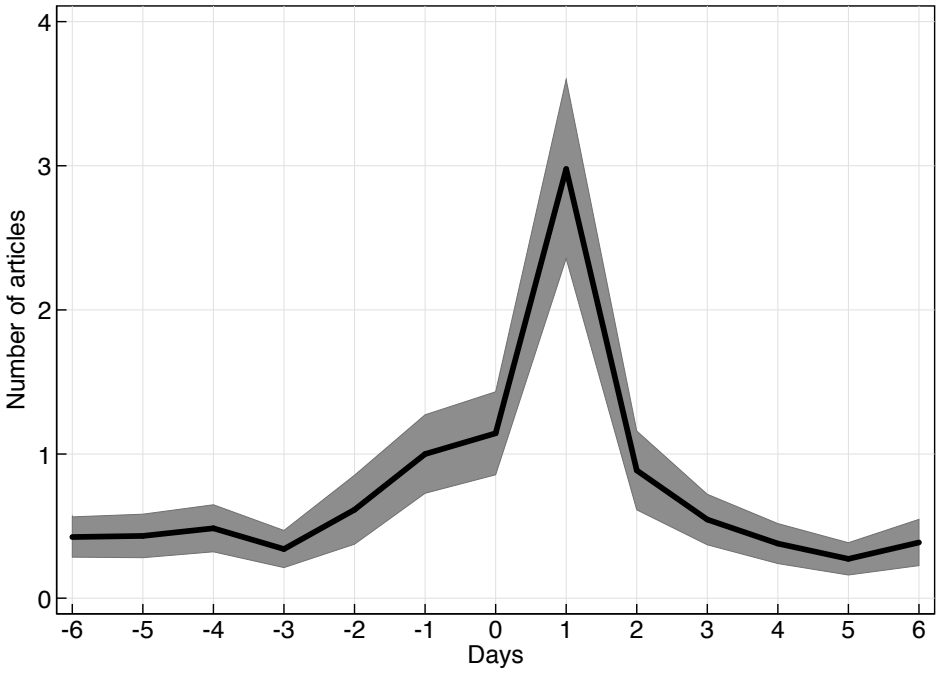


Figure IA.6. Number of Fed-related articles in WSJ&FT around FOMC announcement days. This figure plots the average number of articles that appear in the print issues of the *Wall Street Journal* and the *Financial Times* around days of scheduled FOMC announcements. The gray shaded area shows the two-standard-error deviation bands around the average. The sample period is from February 1994 to March 2011.

Table IA.I Scheduled FOMC Meeting Dates and Times: 1994-2011

This table reports dates of scheduled FOMC meetings. The marker "^{e}" denotes meetings that are excluded from our sample. The time of the announcements, reported in square brackets, from 1994 to 2004 are from Fleming and Piazzesi (2005) and are based on the time-stamp of Bloomberg or Dow Jones newswires. We update this list for the remaining sample using the same method.

Year	1st	2nd	3rd	4th	5th	6th	7th	8th
1994	4-Feb-94^e	22-Mar-94^e	17-May-94^e	6-Jul-94^e	16-Aug-94^e	27-Sep-94	15-Nov-94	20-Dec-94
	[11:05]	[14:20]	[14:26]	[14:18]	[13:17]	[14:18]	[14:20]	[14:17]
1995	1-Feb-95	28-Mar-95	23-May-95	6-Jul-95	22-Aug-95	26-Sep-95	15-Nov-95	19-Dec-95
	[14:14]	[14:13]	[14:13]	[14:15]	[14:13]	[14:14]	[14:16]	[14:15]
1996	31-Jan-96	26-Mar-96^e	21-May-96	3-Jul-96	20-Aug-96	24-Sep-96	13-Nov-96	17-Dec-96
	[14:16]	[11:39]	[14:16]	[14:14]	[14:17]	[14:14]	[14:17]	[14:16]
1997	5-Feb-97	25-Mar-97	20-May-97	2-Jul-97	19-Aug-97	30-Sep-97	12-Nov-97	16-Dec-97
	[14:13]	[14:14]	[14:15]	[14:15]	[14:15]	[14:13]	[14:12]	[14:15]
1998	4-Feb-98	31-Mar-98	19-May-98	1-Jul-98	18-Aug-98	29-Sep-98	17-Nov-98	22-Dec-98
	[14:12]	[14:14]	[14:13]	[14:14]	[14:12]	[14:17]	[14:19]	[14:13]
1999	3-Feb-99	30-Mar-99	18-May-99	30-Jun-99	24-Aug-99	5-Oct-99	16-Nov-99	21-Dec-99
	[14:12]	[14:12]	[14:11]	[14:15]	[14:14]	[14:12]	[14:16]	[14:13]
2000	2-Feb-00	21-Mar-00	16-May-00	28-Jun-00	22-Aug-00	3-Oct-00	15-Nov-00	19-Dec-00
	[14:14]	[14:15]	[14:13]	[14:15]	[14:14]	[14:12]	[14:12]	[14:16]
2001	31-Jan-01	20-Mar-01	15-May-01	27-Jun-01	21-Aug-01	2-Oct-01	6-Nov-01	11-Dec-01
	[14:15]	[14:13]	[14:15]	[14:12]	[14:13]	[14:15]	[14:20]	[14:14]
2002	30-Jan-02	19-Mar-02	7-May-02	26-Jun-02	13-Aug-02	24-Sep-02	6-Nov-02	10-Dec-02
	[14:16]	[14:19]	[14:14]	[14:13]	[14:14]	[14:12]	[14:14]	[14:13]
2003	29-Jan-03	18-Mar-03	6-May-03	25-Jun-03	12-Aug-03	16-Sep-03	28-Oct-03	9-Dec-03
	[14:16]	[14:15]	[14:13]	[14:16]	[14:15]	[14:19]	[14:14]	[14:14]
2004	28-Jan-04	16-Mar-04	4-May-04	30-Jun-04	10-Aug-04	21-Sep-04	10-Nov-04	14-Dec-04
	[14:14]	[14:15]	[14:16]	[14:18]	[14:15]	[14:15]	[14:15]	[14:15]
2005	2-Feb-05	22-Mar-05	3-May-05	30-Jun-05	9-Aug-05	20-Sep-05	1-Nov-05	13-Dec-05
	[14:17]	[14:17]	[14:16]	[14:15]	[14:17]	[14:17]	[14:18]	[14:13]
2006	31-Jan-06	28-Mar-06	10-May-06	29-Jun-06	8-Aug-06	20-Sep-06	25-Oct-06	12-Dec-06
	[14:14]	[14:17]	[14:17]	[14:16]	[14:14]	[14:13]	[14:13]	[14:14]
2007	31-Jan-07	21-Mar-07	9-May-07	28-Jun-07	7-Aug-07	18-Sep-07	31-Oct-07	11-Dec-07
	[14:14]	[14:15]	[14:15]	[14:14]	[14:14]	[14:15]	[14:15]	[14:15]
2008	30-Jan-08	18-Mar-08	30-Apr-08	25-Jun-08	5-Aug-08	16-Sep-08	29-Oct-08	16-Dec-08
	[14:14]	[14:14]	[14:15]	[14:19]	[14:13]	[14:14]	[14:17]	[14:11]
2009	28-Jan-09	18-Mar-09	29-Apr-09	24-Jun-09	12-Aug-09	23-Sep-09	4-Nov-09	16-Dec-09
	[14:14]	[14:17]	[14:16]	[14:18]	[14:16]	[14:16]	[14:18]	[14:15]
2010	27-Jan-10	16-Mar-10	28-Apr-10	23-Jun-10	10-Aug-10	21-Sep-10	3-Nov-10	14-Dec-10
	[14:17]	[14:14]	[14:14]	[14:16]	[14:15]	[14:15]	[14:16]	[14:15]
2011	26-Jan-11	15-Mar-11						
	[14:16]	[14:13]						

Table IA.II Summary Statistics in 1980 Sample S&P500 Excess Returns

This table reports summary statistics for the daily log excess returns on the SPX on FOMC days and non-FOMC days. Standard errors for the mean are reported in square brackets. The sample period is January 2, 1980 to March 31, 2011.

	All Observations		Excl. top/bottom 1%	
	FOMC	nonFOMC	FOMC	nonFOMC
Mean	0.366	0.011	0.34	0.018
	[0.07]	[0.01]	[0.06]	[0.01]
St. Dev.	1.12	1.12	0.9	0.9
Skew	2.31	-1.34	0.37	-0.11
Kurtosis	20.7	36.87	4.19	3.63
Max	9.53	12.06	3.54	2.77
Min	-2.93	-22.91	-2.18	-2.97
Obs.	244	7598	240	7448

Table IA.III CRSP Size Portfolio Regressions: 1994-2011

This table provides results for FOMC dummy variable regressions for different U.S. equity portfolios for the sample period from September 1, 1994 through March 30, 2011. The dependent variables are the daily excess returns on the value-weighted and equally weighted market portfolio from CRSP as well as the 10 size decile portfolios provided on Ken French's website. "FOMC dummy" takes a value of one on days of scheduled FOMC announcements and zero otherwise. "Annual ex-return FOMC" is the average cumulative annual excess return on FOMC days, and "Annual ex-return non-FOMC" is the average cumulative annual excess return on non-FOMC days. "Sharpe ratio" is the annualized Sharpe ratio on pre-FOMC returns.*** significant at 1%, ** significant at 5%, * significant at 10%. Robust standard errors are shown in brackets.

Dependent Variable: %Log Excess Return of CRSP Portfolio Index						
Portfolio:	Value Weighted	Equal Weighted	1st Decile	2nd Decile	3rd Decile	4th Decile
FOMC dummy	0.35	0.25	0.20	0.40	0.42	0.44
	[0.10]***	[0.08]***	[0.08]**	[0.12]***	[0.12]***	[0.11]***
Const.	0.01	0.06	0.02	0.02	0.02	0.01
	[0.02]	[0.02]***	[0.02]	[0.02]	[0.02]	[0.02]
Annual ex-return FOMC	2.90	2.45	1.79	3.32	3.47	3.56
Annual ex-return non-FOMC	2.34	14.66	5.77	3.92	3.69	2.20
Sharpe ratio	0.92	0.93	0.71	0.86	0.93	0.98
Portfolio:	5th Decile	6th Decile	7th Decile	8th Decile	9th Decile	10th Decile
FOMC dummy	0.46	0.40	0.39	0.39	0.37	0.31
	[0.11]***	[0.10]***	[0.10]***	[0.10]***	[0.10]***	[0.10]***
Const.	0.01	0.01	0.02	0.01	0.02	0.01
	[0.02]	[0.02]	[0.02]	[0.02]	[0.02]	[0.02]
Annual ex-return FOMC	3.73	3.26	3.26	3.20	3.10	2.52
Annual ex-return non-FOMC	2.61	3.35	4.23	3.36	3.66	1.80
Sharpe ratio	1.06	1.03	1.05	0.99	0.98	0.80

Table IA.IV CRSP Size Portfolio Regressions: 1980-2011

This table provides results for FOMC dummy variable regressions for different U.S. equity portfolios for the sample period from January 2, 1980 through March 30, 2011. The dependent variables are the daily excess returns on the value-weighted and equally weighted market portfolio from CRSP as well as the 10 size decile portfolios provided on Ken French's website. "FOMC dummy" takes a value of one on days of scheduled FOMC announcements and zero otherwise. "Annual ex-return FOMC" is the average cumulative annual excess return on FOMC days, and "Annual ex-return non-FOMC" is the average cumulative annual excess return on non-FOMC days. "Sharpe ratio" is the annualized Sharpe ratio on pre-FOMC returns.*** significant at 1%, ** significant at 5%, * significant at 10%. Robust standard errors are shown in brackets.

Dependent Variable: %Log Excess Return of CRSP Portfolio Index						
Portfolio:	Value Weighted	Equal Weighted	1st Decile	2nd Decile	3rd Decile	4th Decile
FOMC dummy	0.27	0.13	0.10	0.23	0.26	0.28
	[0.07]***	[0.05]**	[0.05]*	[0.07]***	[0.07]***	[0.07]***
Const.	0.01	0.06	0.02	0.02	0.02	0.02
	[0.01]	[0.01]***	[0.01]*	[0.01]	[0.01]	[0.01]
Annual ex-return FOMC	2.20	1.48	0.89	1.89	2.20	2.28
Annual ex-return non-FOMC	3.42	14.56	4.36	3.79	4.49	3.68
Sharpe ratio	0.78	0.66	0.41	0.60	0.72	0.76
Portfolio:	5th Decile	6th Decile	7th Decile	8th Decile	9th Decile	10th Decile
FOMC dummy	0.29	0.26	0.26	0.27	0.29	0.26
	[0.07]***	[0.06]***	[0.06]***	[0.07]***	[0.07]***	[0.07]***
Const.	0.02	0.02	0.02	0.02	0.02	0.01
	[0.01]	[0.01]	[0.01]*	[0.01]	[0.01]	[0.01]
Annual ex-return FOMC	2.39	2.21	2.22	2.24	2.40	2.15
Annual ex-return non-FOMC	4.38	4.80	5.00	4.53	4.29	3.05
Sharpe ratio	0.81	0.82	0.83	0.78	0.84	0.72

Table IA.V Value-Weighted Industry Portfolio Regressions: 1994-2011

This table provides results for FOMC dummy variable regressions for different U.S. equity portfolios for the sample period from September 1, 1994 to March 30, 2011. The dependent variables are the daily excess returns on the 49 industry portfolios provided on Ken French's website. "FOMC dummy" denotes a variable that takes on the value of one on days of scheduled FOMC announcements and zero otherwise. "Sharpe ratio" is the annualized Sharpe ratio on pre-FOMC returns. *** significant at 1%, ** significant at 5%, * significant at 10%. Robust standard errors are shown in brackets.

Dependent Variable:	%Log Excess Return of CRSP Industry Portfolio						
Industry Portfolio:	AGRIC	FOOD	SODA	BEER	SMOKE	TOYS	FUN
FOMC dummy	0.24	0.10	-0.14	0.01	-0.07	0.32	0.50
	[0.17]	[0.08]	[0.22]	[0.09]	[0.15]	[0.12]**	[0.19]***
Sharpe ratio	0.39	0.38	-0.10	0.09	-0.06	0.61	0.66
Industry Portfolio:	BOOKS	HSHLD	CLTHS	HLTH	MEDEQ	DRUGS	CHEMS
FOMC dummy	0.37	0.23	0.38	0.24	0.19	0.25	0.44
	[0.12]***	[0.09]**	[0.13]***	[0.11]**	[0.10]*	[0.10]***	[0.12]***
Sharpe ratio	0.73	0.67	0.77	0.57	0.54	0.72	0.98
Industry Portfolio:	RUBBR	TXTLS	BLDMT	CNSTR	STEEL	FABPR	MACH
FOMC dummy	0.39	0.51	0.38	0.44	0.50	0.51	0.49
	[0.12]***	[0.16]***	[0.13]***	[0.18]**	[0.16]***	[0.19]***	[0.13]***
Sharpe ratio	0.80	0.77	0.70	0.63	0.78	0.63	0.97
Industry Portfolio:	ELCEQ	AUTOS	AERO	SHIPS	GUNS	GOLD	MINES
FOMC dummy	0.53	0.51	0.45	0.25	0.13	0.26	0.56
	[0.14]***	[0.14]***	[0.15]***	[0.14]*	[0.13]	[0.22]	[0.16]***
Sharpe ratio	0.98	0.86	0.79	0.49	0.29	0.28	0.89
Industry Portfolio:	COAL	OIL	UTIL	TELCM	PERSV	BUSSV	HARDW
FOMC dummy	0.40	0.29	0.10	0.13	0.42	0.37	0.40
	[0.27]	[0.12]**	[0.09]	[0.12]	[0.13]***	[0.10]***	[0.17]**
Sharpe ratio	0.43	0.67	0.34	0.28	0.82	0.93	0.63
Industry Portfolio:	SOFTW	CHIPS	LABEQ	PAPER	BOXES	TRANS	WHLSL
FOMC dummy	0.48	0.54	0.44	0.18	0.27	0.45	0.30
	[0.15]***	[0.16]***	[0.14]***	[0.10]*	[0.12]**	[0.11]***	[0.10]***
Sharpe ratio	0.80	0.82	0.83	0.46	0.57	0.99	0.76
Industry Portfolio:	RTAIL	MEALS	BANKS	INSUR	RLEST	FIN	OTHER
FOMC dummy	0.33	0.27	0.56	0.42	0.35	0.70	0.30
	[0.12]***	[0.11]**	[0.19]***	[0.14]***	[0.16]**	[0.19]***	[0.12]***
Sharpe ratio	0.74	0.64	0.72	0.77	0.53	0.92	0.61

Table IA.VI CRSP Value-Weighted Industry Portfolio Regressions: 1980-2011

This table provides results for FOMC dummy variable regressions for different U.S. equity portfolios for the sample period from January 2, 1980 to March 30, 2011. The dependent variables are the daily excess returns on the 49 industry portfolios provided on Ken French's website. "FOMC dummy" denotes a variable that takes on the value of one on days of scheduled FOMC announcements and zero otherwise. "Sharpe ratio" is the annualized Sharpe ratio on pre-FOMC returns. *** significant at 1%, ** significant at 5%, * significant at 10%. Robust standard errors are shown in brackets.

Dependent Variable:	%Log Excess Return of CRSP Industry Portfolio						
Industry Portfolio:	AGRIC	FOOD	SODA	BEER	SMOKE	TOYS	FUN
FOMC dummy	0.12	0.14	-0.02	0.08	0.09	0.25	0.33
	[0.10]	[0.06]**	[0.13]	[0.08]	[0.10]	[0.09]***	[0.12]***
Sharpe ratio	0.26	0.53	0.02	0.29	0.24	0.49	0.55
Industry Portfolio:	BOOKS	HSHLD	CLTHS	HLTH	MEDEQ	DRUGS	CHEMS
FOMC dummy	0.27	0.20	0.28	0.25	0.17	0.30	0.30
	[0.08]***	[0.07]***	[0.08]***	[0.08]***	[0.07]**	[0.07]***	[0.08]***
Sharpe ratio	0.65	0.60	0.65	0.62	0.48	0.82	0.75
Industry Portfolio:	RUBBR	TXTLS	BLDMT	CNSTR	STEEL	FABPR	MACH
FOMC dummy	0.22	0.37	0.25	0.28	0.31	0.40	0.30
	[0.08]***	[0.09]***	[0.08]***	[0.11]**	[0.10]***	[0.11]***	[0.08]***
Sharpe ratio	0.53	0.73	0.58	0.48	0.58	0.63	0.69
Industry Portfolio:	ELCEQ	AUTOS	AERO	SHIPS	GUNS	GOLD	MINES
FOMC dummy	0.39	0.35	0.32	0.21	0.18	0.29	0.33
	[0.09]***	[0.10]***	[0.10]***	[0.10]**	[0.09]**	[0.15]*	[0.10]***
Sharpe ratio	0.81	0.67	0.66	0.38	0.39	0.34	0.59
Industry Portfolio:	COAL	OIL	UTIL	TELCM	PERSV	BUSSV	HARDW
FOMC dummy	0.33	0.22	0.11	0.10	0.26	0.25	0.30
	[0.16]**	[0.09]**	[0.06]**	[0.08]	[0.08]***	[0.07]***	[0.11]***
Sharpe ratio	0.40	0.51	0.43	0.28	0.57	0.72	0.52
Industry Portfolio:	SOFTW	CHIPS	LABEQ	PAPER	BOXES	TRANS	WHLSL
FOMC dummy	0.33	0.38	0.30	0.16	0.23	0.34	0.24
	[0.11]***	[0.10]***	[0.10]***	[0.07]**	[0.08]***	[0.08]***	[0.07]***
Sharpe ratio	0.56	0.71	0.60	0.47	0.55	0.80	0.68
Industry Portfolio:	RTAIL	MEALS	BANKS	INSUR	RLEST	FIN	OTHER
FOMC dummy	0.24	0.22	0.32	0.28	0.15	0.47	0.25
	[0.08]***	[0.07]***	[0.11]***	[0.08]***	[0.10]	[0.11]***	[0.08]***
Sharpe ratio	0.62	0.60	0.54	0.63	0.25	0.80	0.56

References

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